A

Internship Report on

Counting Number of Boxes in a Warehouse using Computer Vision

Submitted in partial fulfilment of the requirements for the award of Degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

By

Praneethanjanee Karanam(160622733092)

Under the Guidance of

Mrs. Shaheda Begum

Assistant Professor
Department of Computer Science and Engineering

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
STANLEY COLLEGE OF ENGINEERING & TECHNOLOGY FOR WOMEN
(AUTONOMOUS)
(Affiliated to Osmania University, Approved by AICTE and Accredited by NAAC)
Hyderabad - 500 001, Telangana, India

STANLEY COLLEGE OF ENGINEERING & TECHNOLOGY FOR WOMEN

(AUTONOMOUS)

(Affiliated to Osmania University, Approved by AICTE and Accredited by NAAC)

Hyderabad - 500 001, Telangana, India

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the Internship report entitled, **"Counting Number of Boxes in a Warehouse using Computer Vision"**, done by **Praneethanjanee Karanam (160622733092),**
Submitting to the Department of Computer Science and Engineering, **STANLEY COLLEGE OF ENGINEERING & TECHNOLOGY FOR WOMEN**, in partial fulfilment of the requirements for the Degree of **BACHELOR OF ENGINEERING** in **Computer Science and Engineering**, **Purview India Consulting and Services LLP** during the year 2024-25. It is certified that she has completed the project satisfactorily.

Signature of Supervisor / Faculty Coordinator	**Signature of Head of the Department**
Mrs. Shaheda Begum	**Dr. R. Madana Mohana**
Assistant Professor	Professor & Head

STANLEY COLLEGE OF ENGINEERING & TECHNOLOGY FOR WOMEN

(AUTONOMOUS)

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Hyderabad - 500 001, Telangana, India

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

DECLARATION

I hereby declare that the work described in this report entitled **"Counting Number of Boxes in a Warehouse using Computer Vision"** which is being submitted by us in partial fulfilment for the award of **BACHELOR OF ENGINEERING** in the Department of Computer Science and Engineering, Stanley College of Engineering & Technology for Women to the Osmania University Hyderabad.

The work is original and has not been submitted for any Degree or Diploma of this or any other university.

Signature of the Student

Praneethanjanee Karanam
(160622733092)

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Hyderabad - 500 001, Telangana, India

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

INTERNSHIP - II DETAILS

Course Code: SPW611 CS   Credits: 1   Evaluation: CIE - 50 Marks

Title: Counting Number of Boxes in a Warehouse using Computer Vision

Name of the Student: Praneethanjanee Karanam

Roll Number: 160622733092

Batch / Section: 2024 – 2025 / B

Name of the Organization: Purview India Consulting and Services LLP

Duration of Internship: From 02 May 2025 To 16 June 2025

Mode (Onsite / Online / Hybrid): Online

Industry Supervisor: Dr. Manoj Patil

Faculty Coordinator / Supervisor: Mrs. Shaheda Begum

ORGANIZATION PROFILE

Overview of the Company

Purview India Consulting and Services LLP is the Indian-arm of the global Purview Services group, offering IT services, consulting and digital transformation solutions from its Hyderabad office located at Ground Floor, Building H03, 34P-35P, Gachibowli Rd., Phase 2, HITEC City, Telangana-500081. The company supports clients by integrating modern technologies across software, infrastructure and business-process domains.

Vision, Mission, and Key Areas of Operation

Vision: To empower organisations through technology innovation so they stay competitive and future-ready.
Mission: To deliver scalable, end-to-end digital services including AI/ML, cloud, analytics and automation that drive operational excellence and business value.

Key Areas of Operation:

- Artificial Intelligence & Machine Learning (including image processing and computer vision)
- Data, BI & Analytics
- Cloud Services & DevOps
- Internet of Things (IoT), Digital Transformation and Automation
- Cyber Security and Managed IT Services

Organizational Structure

The organization follows a lean and agile project-delivery structure:

- Managing Partners / Directors oversee strategic operations
- Project Managers coordinate cross-functional teams

- Technical Leads guide the software/data/AI teams
- Developers, Engineers and Interns execute the implementation, testing and deployment phases

Technologies or Tools Used

Purview Services' technology stack includes:

- **Programming Languages:** Python, Java, JavaScript
- **Frameworks/Platforms:** OpenCV (for computer vision), PyTorch/TensorFlow (AI/ML), React/Flask (frontend/backend)
- **Cloud & DevOps Tools:** AWS, Git/GitHub, Docker, Kubernetes
- **Data & Analytics:** MySQL/MongoDB, Data-Lakes, BI dashboards
- **Automation & IoT Devices:** Sensor networks, image/video capture systems, edge devices for deployment

Internship Offer Letter

PURVIEW

Purview India Consulting & Services LLP
3rd Floor, Trendz Solitaire,
Near The Westin Hotel, Gafoor Nagar
Hitechcity, Hyderabad - 500 081.
LLP IN: AAG-04202

01st May 2025

Internship Offer Letter

Dear **Praneethanjanee Karanam,**

We are pleased to inform your acceptance of an internship position as **Software Trainee** at Purview India Consulting and services LLP and Date of joining is on **02/05/2025**.

Duration: **45 days**

During which, you will have access to the company's clients and confidential information. You will not share this information with anyone outside the company and not use it for your benefits.

Congratulations! Welcome Onboard.

For Purview India Consulting and services LLP

Authorized Signatory

hr@purviewservices.com
www.purviewservices.com

Our Branches:

Internship Completion Certificate

PURVIEW

Purview India Consulting & Services LLP
3rd Floor, Trendz Solitaire,
Near The Westin Hotel, Gafoor Nagar
Hitechcity, Hyderabad - 500 081.
LLP IN: AAG-04202

18th July 2025

Internship Completion Letter

Dear **Praneethanjanee Karanam,**

We are pleased to confirm that you have successfully completed your internship with Purview India Consulting Services LLP from **02-May-2025** to **16-June-2025**.

During your time with us, you have demonstrated a high level of dedication, professionalism, and skill in your role as a **Software Trainee** on project **Counting Number of Boxes in a Warehouse Using Computer Vision**.

We wish you all the best in your future endeavors and are confident that you will continue to achieve great success.

For Purview India Consulting and Services LLP

Authorized Signatory

hr@purviewservices.com
www.purviewservices.com

Our Branches:

Internship Mentoring Appreciation Certificate

PURVIEW

Purview India Consulting & Services LLP
3rd Floor, Trendz Solitaire,
Near The Westin Hotel, Gafoor Nagar
Hitechcity, Hyderabad - 500 081.
LLP IN: AAG-04202

18th July 2025

To
M Sara Angeline.
Computer Science and Engineering Department,
Stanley college of engineering and technology for women.

Subject: Appreciation for Mentoring Students During Industry Internship

Dear **M Sara Angeline**,

We extend our heartfelt appreciation for your valuable mentorship and guidance provided to interns during their industry internship program conducted from **02/05/2025** to **16/06/2025**.

Your dedication and consistent support have played a pivotal role in helping interns bridge the gap between academic learning and practical exposure. Your mentorship reflects the quality of insights and professionalism.

We sincerely thank you for your contribution to the overall growth and development of interns and look forward to your continued involvement in future academic–industry collaboration initiatives.

Interns details:

Sr. No.	List of interns	Internship Project Title
1	Anusha Adula	Counting Number of Boxes in a Warehouse Using Computer Vision
2	Ganja Srividya	
3	Vemulavada Sudheshna	
4	Aishwarya Pallanti	
5	Nukala Manasa	
6	Praneethanjanee Karanam	

For Purview India Consulting and Services LLP

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Our Branches:

Internship Mentoring Appreciation Certificate

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Near The Westin Hotel, Gafoor Nagar
Hitechcity, Hyderabad - 500 081.
LLP IN: AAG-04202

18th July 2025

To
B. Soujanya.
Computer Science and Engineering Department,
Stanley college of engineering and technology for women.

Subject: Appreciation for Mentoring Students During Industry Internship

Dear **B. Soujanya**,

We extend our heartfelt appreciation for your valuable mentorship and guidance provided to interns during their industry internship program conducted from **02/05/2025** to **16/06/2025**.

Your dedication and consistent support have played a pivotal role in helping interns bridge the gap between academic learning and practical exposure. Your mentorship reflects the quality of insights and professionalism.

We sincerely thank you for your contribution to the overall growth and development of interns and look forward to your continued involvement in future academic–industry collaboration initiatives.

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4	Aishwarya Pallanti	
5	Nukala Manasa	
6	Praneethanjanee Karanam	

For Purview India Consulting and Services LLP

Authorized Signatory

hr@purviewservices.com
www.purviewservices.com

Our Branches:

Department of Computer Science and Engineering

Vision of the Institute

Empowering girl students through professional education integrated with values and character to make an impact in the world.

Mission of the Institute

- Providing quality engineering education for girl students to make them competent and confident to succeed in professional practice and advanced learning.
- Establish state-of-art-facilities and resources to facilitate world class education.
- Integrating qualities like humanity, social values, ethics, leadership in order to encourage contribution to society.

Vision of the Department

Empowering girl students with the contemporary knowledge in computer science engineering for their success in life.

Mission of the Department

- To impart quality education for girl students to learn and practice various hardware and software platforms prevalent in industry.
- To achieve self-sustainability and overall development through Research and Development activities.
- To provide education for life by focusing on the inculcation of human & moral values through an honest and scientific approach.
- To groom students with good attitude, team work and personality skills.

Program Educational Objectives (PEOs)

PEO1. Graduates shall have enhanced skills and contemporary knowledge in software and hardware technologies for professional excellence, towards successful employment, advanced learning and research.

PEO2. Graduates shall have life-long learning attitude, innovation and creativity to master latest technologies, devise solutions for realistic and social issues in the society.

PEO3. Graduates shall have good attitude and personality skills, ethical values, teamwork and leadership skill towards professionalism and ethical practices within the organization and the society.

Department of Computer Science and Engineering

Knowledge and Attitude Profile (WK)

WK1.	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
WK2.	Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
WK3.	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
WK4.	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
WK5.	Knowledge, including efficient resource use, environmental impacts, whole-life cost, reuse of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
WK6.	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
WK7.	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.
WK8.	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.
WK9.	Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

Department of Computer Science and Engineering

Program Outcomes (POs)

PO1.	**Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
PO2.	**Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
PO3.	**Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
PO4.	**Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
PO5.	**Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
PO6.	**The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
PO7.	**Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
PO8.	**Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
PO9.	**Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
PO10.	**Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
PO11.	**Life-Long Learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

Program Specific Outcomes (PSOs)

PSO1.	**Problem-Solving Skills**: The ability to apply standard practices and strategies in software project development using open-ended programming environments to deliver a quality product for the benefit of students.
PSO2.	Design, Implement, Test and Evaluate a computer system, component or algorithm to meet desired needs and to solve a computational problem.

Objectives and Outcomes of the Internship

Course Objectives:

1. Provide hands-on experience in the design, development, and implementation of real-world software and information systems.
2. Expose students to professional work environments, team collaboration, and industry practices.
3. Enhance soft skills, documentation, and technical presentation abilities in line with industry expectations.
4. Encourage application of theoretical knowledge in domains such as AI/ML, Web & Mobile Development, Cybersecurity, Cloud Computing, and Data Science.
5. Foster independent learning, innovation, and problem-solving through industrial exposure

Course Outcomes:

After completing this course, the student will be able to:

1. Gain practical experience in software development, coding standards, and tools used in industrial or R&D settings.
2. Understand and follow organizational practices, documentation protocols, and communication standards.
3. Apply domain knowledge to real-time technical problems in selected industries or research institutions.
4. Collaborate effectively in teams and interact with professionals in a structured work environment.
5. Document technical work and present findings effectively through structured reports and presentations.

Mapping of Course Outcomes with POs and PSOs

Course Outcomes (COs)	Program Outcomes (POs)											Program Specific Outcomes (PSOs)	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
Gain practical experience in software development, coding standards, and tools used in industrial or R&D settings.	3	3	3	3	3	2	-	-	2	2	3	3	3
Understand and follow organizational practices, documentation protocols, and communication standards.	2	2	-	2	2	2	2	2	3	3	3	2	2
Apply domain knowledge to realtime technical problems in selected industries or research institutions.	3	3	3	3	3	3	-	-	2	2	3	3	2
Collaborate effectively in teams and interact with professionals in a structured work environment.	-	-	-	-	2	-	2	3	3	2	2	2	-
Document technical work and present findings effectively through structured reports and presentations.	-	2	-	2	-	-	-	3	3	3	3	2	-

Student Learning Outcomes

Skills Acquired :

During the internship at Purview India Consulting and Services LLP, I gained handson experience in computer vision and image processing applications for real-world industrial use.

- **Technical Skills:** Learned to design and implement object detection and counting algorithms using deep learning frameworks. Improved my understanding of dataset preparation, image annotation, and model evaluation metrics.

- **Analytical Skills:** Enhanced my ability to analyze visual data, fine-tune models for accuracy, and interpret system outputs in relation to physical warehouse conditions.

- **Communication & Teamwork:** Collaborated with mentors and peers to discuss algorithm improvements and documentation. Improved professional communication through progress reporting and presentation of results.

Tools / Technologies Learned:

- Worked with **Python**, **OpenCV**, and **TensorFlow** for developing and training object detection models.

- Used **LabelImg** for data annotation and **NumPy**, **Pandas**, and **Matplotlib** for data analysis and visualization.

- Understood integration of the trained model with real-time camera feeds and optimization for industrial deployment.

Industry Exposure and Professional Ethics:

This internship provided valuable exposure to how computer vision solutions are applied in logistics and inventory automation. I learned the importance of accuracy, efficiency, and reliability in industrial AI applications. It also reinforced professional ethics such as responsible data handling, maintaining confidentiality of warehouse operations, teamwork discipline, and adhering to project timelines.

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Department of Computer Science and Engineering

Mapping of Internship with SDGs

Roll No: 160622733092
Name: Praneethanjanee Karanam
Class, Semester & Section: 3, VI, B
Academic Year: 2024 - 2025
Title of Internship: Counting Number of Boxes in a Warehouse Using Computer Vision
Domain: Artificial Intelligence

Course Outcomes (COs) Mapping with SDGs:

SDG Mapping with Justification

Course Outcomes (COs)	Relevant SDGs	Justification of SDGs
CO1: Gain practical experience in software development, coding standards, and tools used in industrial or R&D settings.	**SDG 4 – Quality Education** **SDG 9 – Industry, Innovation and Infrastructure**	The internship enhanced practical learning through AIdriven warehouse automation, aligning with SDG 4 by bridging academic knowledge with industrial application. It also supports SDG 9 by fostering innovation and improving industrial infrastructure through automation.
CO2: Understand and follow organizational practices, documentation protocols, and communication standards.	**SDG 8 – Decent Work and Economic Growth** **SDG 16 – Peace, Justice and Strong Institutions**	Following ethical standards, maintaining data confidentiality, and ensuring fair AI use promote a responsible and sustainable work culture (SDG 8) while strengthening institutional trust and ethical integrity (SDG 16).
CO3: Apply domain knowledge to realtime technical problems in selected industries or research institutions.	**SDG 9 – Industry, Innovation and Infrastructure** **SDG 12 – Responsible Consumption and Production**	The automation of box counting enhances operational efficiency and reduces manual errors, directly contributing to industrial innovation (SDG 9) and promoting efficient resource usage and sustainable production (SDG 12)..
CO4: Collaborate effectively in teams and interact with	**SDG 5 – Gender Equality** **SDG 17 – Partnerships for the Goals**	The project involved equal contribution and teamwork among all members, fostering

professionals in a structured work environment.		inclusivity (SDG 5) and collaboration aligned with global partnership goals (SDG 17).
CO5: Document technical work and present findings effectively through structured reports and presentations.	**SDG 4 – Quality Education** **SDG 12 – Responsible Consumption and Production**	Structured documentation and clear presentation enhance knowledge sharing (SDG 4) and encourage professional, ethical communication practices (SDG 12).

Internship Domain Mapping with SDGs:

Internship Domain	Relevant SDGs	Justification
Computer Vision & AI for Warehouse Automation	**SDG 9 - Industry, Innovation and Infrastructure** **SDG 12 – Responsible Consumption and Production**	Designed and implemented an automated box-counting system using computer vision to improve industrial accuracy and reduce operational waste, directly supporting innovation in logistics infrastructure (SDG 9) and responsible, efficient production systems (SDG 12).

ACKNOWLEDGEMENT

The satisfaction that accompanies the successful completion of the task would be put incomplete without the mention of the people who made it possible, whose constant guidance and encouragement crown all the efforts with success.

The organization and our mentors **B. Soujanya and M. Sara Angeline** are very helpful and continuously encourage me to complete the task assigned by the company. Organization support is also very good. They provided all the basic requirements required to complete the task.

We express our heartfelt thanks to **Mrs. Shaheeda Begum**, Assistant Professor, for her suggestions in the selection and carrying out an in-depth study of the topic. Her valuable guidance and encouragement really helped us to shape this report to perfection.

We express our heartfelt thanks to **Mr. Manoj Patil** for his suggestions invaluable inputs and assessment really helped us to shape this report to perfection.

We wish to express our deep sense of gratitude to **Dr. R. Madana Mohana**, Professor & Head, for his intense support and encouragement, which helped us to mold our internship into a successful one.

We also owe our special thanks to our honorable Principal **Dr. B. L. Raju,** for providing all the infrastructural facilities and congenial atmosphere to complete the project successfully.

We also thank all the staff members of the Computer Science and E department for their valuable support and generous advice.

Finally, thanks to all my friends and family members for their continuous support and enthusiastic help.

Praneethanjanee Karanam
(160622733092)

ABSTRACT

The internship at **Purview India Consulting and Services LLP** Hyderabad, focused on developing an automated computer vision solution to **count the number of boxes in a warehouse** environment. The primary objective was to design a reliable system capable of detecting and quantifying stacked boxes from camera-captured images or video feeds, thereby reducing manual effort and improving inventory accuracy.

The project involved implementing object detection algorithms using **Python**, **OpenCV**, and **TensorFlow**, along with dataset preparation, annotation, and model training. Emphasis was placed on optimizing detection accuracy under varying lighting and positioning conditions commonly found in industrial settings.

Through this internship, I gained practical exposure to machine learning workflows, image processing techniques, and model evaluation metrics. Additionally, the experience enhanced my understanding of project documentation, teamwork, and problem-solving in a professional software environment. The outcome of the project demonstrated a functional prototype capable of assisting warehouse automation systems by efficiently identifying and counting boxes in real time.

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

SYMBOLS & ABBREVIATIONS

Abbreviation	Meaning	PAGE NO.
AI	Artificial Intelligence	1
RFID	Radio Frequency Identification	2
YOLO	You Only Look Once	2
R-CNN	Region based Convolutional Neural Networks	3
CV	Computer Vision	3
GUI	Graphical User Interface	2
PIL	Pillow (python library)	3
mAP	Mean Average Precision	8
IoU	Intersection Over Union	8

WORK / PROJECT DESCRIPTION

1. **Problem statement**
2. **Background and literature / technology review**
3. **Planning and design of work**
4. **Implementation / execution**
5. **Testing / validation**
6. **Results and analysis**
7. **Challenges and Solutions**
8. **Conclusion and Future Scope**

1. PROBLEM STATEMENT

The main problem addressed during the internship at **Purview India Consulting and Services LLP** was the **manual counting of boxes in warehouse environments**, which was time-consuming, prone to human error, and inefficient for large-scale operations. Accurate inventory tracking is critical for supply chain management, and errors in counting can lead to stock mismatches, delays, and financial losses.

The objective of the project was to **develop an automated box counting system using computer vision techniques** that could accurately detect and count boxes from real-time images or video feeds. The system aimed to minimize manual intervention, improve operational accuracy, and enhance productivity within warehouse management systems.

This automation solution was designed to support Purview's goal of leveraging **AI and computer vision** technologies to deliver intelligent, scalable solutions for industrial automation and digital transformation.

2. BACKGROUND AND LITERATURE / TECHNOLOGY REVIEW

2.1 Background & Context

Warehouse inventory management is a critical component of supply chain operations. Traditionally, box counting and stock verification are performed manually by staff, which often leads to **inaccuracies, delays, and labor overhead**. With warehouses scaling larger and adopting automation systems, the need for **real-time, reliable object counting** has become essential.

Earlier inventory tracking methods relied on **barcode scanners or RFID tags**, requiring labels to be printed and attached to boxes. While accurate, these processes are timeconsuming and depend on physical involvement. This paved the way for **computer vision– based automation**, where **cameras and AI** can automatically detect and count items without manual tagging.

Modern **deep learning models** such as YOLO, Faster R-CNN, and EfficientDet have shown the ability to detect objects in complex environments with high accuracy. By integrating these models with **OpenCV for visual processing and GUI tools such as Tkinter**, industries can build practical, user-friendly automation systems.

This project aligns with these industrial needs by developing a **real-time box counting system** that uses **YOLO-based object detection** and a **Tkinter-based interface**, allowing warehouse staff to monitor box count efficiently.

2.2 Literature / Research Review

S.No.	Paper / Source	Author(s) & Year	Key Contribution / Findings	Relevance to Project
1	You Only Look Once: Unified, Real-Time Object Detection	Joseph Redmon et al., 2016	Proposed the original YOLO framework that performs object detection in a single forward pass, enabling real-time inference.	Provided the conceptual foundation for the object detection system implemented in this project.
2	YOLOv8 Technical Documentation	Glenn Jocher, Ultralytics, 2023	Detailed improvements such as anchor-free design, hybrid backbone, and mosaic data augmentation.	Guided model training, dataset preparation, and optimization process.
3	Faster R-CNN: Towards RealTime Object Detection with Region Proposal Networks	Shaoqing Ren et al., 2015	Introduced region proposal networks (RPNs) for improved detection accuracy and localization.	Compared during evaluation; too computationally heavy for realtime use.
4	EfficientDet: Scalable and Efficient Object Detection	Mingxing Tan et al., 2019	Proposed compound scaling to balance model complexity and performance.	Used for baseline comparison; underperformed in dense warehouse scenarios.
5	Detectron2: A Modular Object Detection Framework	Facebook AI Research, 2021	Presented a flexible research framework supporting Faster RCNN, Mask R-CNN, and RetinaNet.	Provided reference architecture; helpedbenchmark YOLOv8's simplicity and deployment ease.
6	Deep Learning for Warehouse Automation: A Review	Zhang, Li & Liu, IEEE Access, 2020	Comprehensive review of AI and computer vision applications in warehouse management and robotics.	Strengthened understanding of how AI-driven detection improves logistics automation.

7	Real-Time Object Detection for Industrial Automation using YOLOv5	Ahmed et al., International Journal of Advanced Computer Science, 2021	Showed successful realtime object counting in production lines using YOLOv5.	Provided comparative insights into YOLO-based systems for object counting tasks.
8	A Comparative Study of Object Detection Algorithms for Visual Inspection Systems	Patel & Shah, Springer, 2022	Compared YOLO, SSD, and R-CNN models for real-time industrial use.	Helped validate why YOLOv8 was best suited for box detection in cluttered environments.
9	Applications of Computer Vision in Inventory and Logistics Management	Singh et al., Elsevier Procedia Computer Science, 2021	Demonstrated camerabased automation for real-time inventory verification.	Supported the relevance of computer vision as a scalable warehouse management tool.
10	Automated Counting and Detection of Packages Using Deep Neural Networks	Kumar & Das, IEEE ICACCP, 2022	Focused on automated package detection using CNN architectures.	Reinforced our approach of using deep learning for accurate box counting and tracking.

2.3 Technologies and Tools Used

Technology / Tool	Category	Purpose / Role in Project
Python 3.x	Programming Language	Used to develop the overall application including model integration, image processing, and GUI development.
YOLOv8 (Ultralytics)	Object Detection Model	Detects and localizes boxes in each frame using deep learning, providing bounding boxes and confidence scores.
OpenCV	Computer Vision Library	Handles video input, frame-by-frame processing, drawing bounding boxes, and image transformations.
PyTorch	Deep Learning Framework	Backend framework used to load and run YOLOv8 weights efficiently.
Tkinter	GUI Framework (Python Standard Library)	Creates a clean desktop interface for user controls like upload, live camera feed, and display results.
Pillow (PIL)	Image Handling Library	Converts OpenCV frames to GUI-compatible image formats for display.
Webcam / Video Feed	Hardware Input Source	Captures live video stream for real-time box detection and counting demonstration.

3. PLANNING AND DESIGN OF WORK

The internship was systematically planned and executed over a six-week duration, from **May to mid-June 2025**, to ensure progressive development of both technical and analytical skills. The project titled **“Counting Number of Boxes in a Warehouse using Computer Vision”** focused on building an intelligent system capable of detecting and counting boxes using image processing and deep learning techniques.

3.1 Internship Planning and Timeline

Week	Tasks
Week 1	Understood the problem statement and objectives of warehouse automation. Conducted initial research on computer vision, object detection algorithms, and YOLO models. Installed required tools and set up the working environment (Python, OpenCV, YOLOv8, Tkinter).
Week 2	Collected and annotated dataset images of warehouse boxes under varying lighting and orientation conditions. Used **LabelImg** to annotate the dataset and divided it into training and validation sets.
Week 3	Studied and evaluated different object detection models such as YOLOv11, RetinaNet, and Faster RCNN. Selected **YOLOv8** based on high performance in precision, recall, and mean Average Precision (mAP)
Week 4	Trained and fine-tuned the YOLOv8 model using the custom dataset. Adjusted parameters like learning rate, batch size, and number of epochs for improved accuracy and reduced false detections.
Week 5	Integrated the trained model with a **Tkinter GUI** interface to enable real-time box counting from webcam input or uploaded images. Implemented user interaction features such as live preview and result saving.
Week 6	Conducted final testing, accuracy validation, and documentation. Prepared charts, tables, and system diagrams for report inclusion. Compiled results and finalized conclusions for submission.

3.2 Workflow Design

The workflow for the internship project consisted of multiple stages designed to ensure smooth progress and measurable results:

Data Collection → Annotation → Model Training → Testing and Validation → GUI Integration → Result Analysis

This systematic structure ensured an incremental and iterative approach to development.

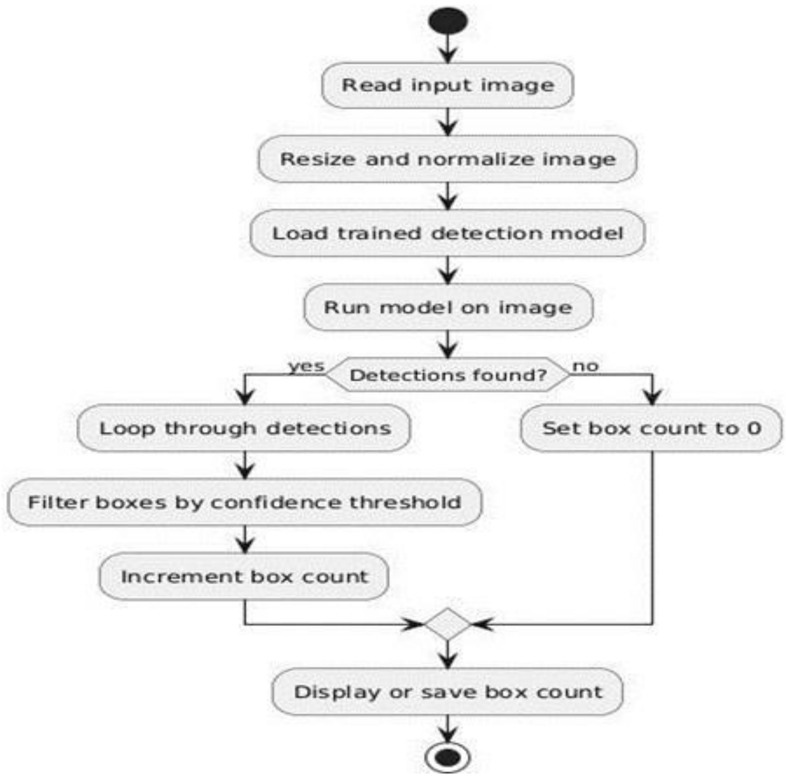

Figure 3.1: Workflow Diagram of the System

3.3 System Architecture

The system was designed in a modular architecture to ensure flexibility, scalability, and easeof maintenance. The main components of the system include:

1. **Input Module:** Captures real-time images or video feed from the webcam or through manual upload.
2. **Pre-processing Module:** Normalizes and resizes the images to meet the YOLOv8 model input requirements.
3. **Detection Module:** Performs object detection using the YOLOv8 algorithm and draws bounding boxes around detected boxes.
4. **Counting Module:** Computes and displays the total count of detected boxes on the GUI.
5. **Output Module:** Visualizes detection results and enables users to save the output frames.

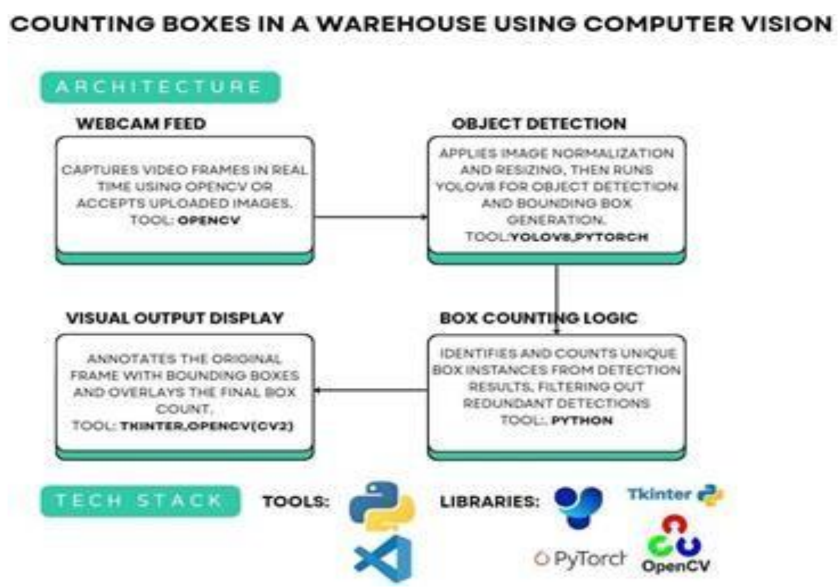

Figure 3.2: System Architecture Diagram

4. IMPLEMENTATION/EXECUTION

4.1 Overview

The project **"Counting Boxes in a Warehouse Using Computer Vision"** was implemented using Python-based tools and frameworks to automate the detection and counting of boxes in warehouse environments. The main goal was to replace manual counting methods with a realtime, intelligent system capable of identifying boxes through video input or static images. The solution integrates deep learning, computer vision, and a graphical user interface, ensuring high accuracy and ease of use for warehouse monitoring applications.

4.2 Tools and Technologies

The system was developed using **Python** as the core programming language, supported by libraries such as **OpenCV**, **Tkinter**, **PyTorch**, and **PIL**. **YOLOv8 (Ultralytics)** served as the object detection model due to its superior accuracy and real-time performance capabilities. OpenCV was used for image and video frame processing, Tkinter for building the desktopbased GUI, PyTorch for model integration and training, and PIL for image handling within the interface. These technologies were combined to form a modular and scalable architecture that allows users to easily interact with the system.

4.3 Dataset Preparation

A **custom dataset** of **1049 images** was created to accurately represent real warehouse conditions. The images were collected manually under various lighting, angle, and background conditions to ensure model robustness. Each image was annotated manually with bounding boxes marking the visible boxes, and the annotations were exported in both **YOLO** and **COCO** formats. The dataset was divided into **80% for training (839 images)** and **20% for validation (210 images)**. This dataset provided a solid foundation for model training and evaluation.

4.4 Model Training and Selection

To identify the most suitable model, several object detection architectures—**YOLOv8**, **YOLOv11**, **RetinaNet**, **Faster R-CNN**, **EfficientDet-D0**, and **Tiny YOLOv3**—were tested and compared using evaluation metrics such as **Precision**, **Recall**, and **Mean Average Precision (mAP@0.5 and mAP@0.5:0.95)**. After analysis, **YOLOv8** was selected as the final model due to its optimal balance between speed and detection accuracy. The model was finetuned using the **yolov8m.pt** base weights for **100 epochs**, resulting in strong generalization and high mAP performance.

4.5 System Integration

The final system integrated three main modules — **camera capture**, **YOLOv8 detection**, and **Tkinter-based GUI**. The OpenCV module managed live video feed and frame capture, the YOLOv8 module performed detection and box counting, and the GUI module provided an interactive interface for users to upload images, start live detection, and view or save outputs. The GUI was designed to be intuitive and responsive, ensuring smooth operation during realtime detection. The overall system was structured modularly, allowing easy updates or future expansions such as multi-object detection or edge device deployment.

4.6 Summary

In summary, the implementation successfully combined **deep learning**, **computer vision**, and **user interface design** into a complete, real-time warehouse box-counting solution. The use of YOLOv8 ensured fast and accurate detection, while the Tkinter-based interface made the system accessible and user-friendly. The entire implementation demonstrates how AI can be practically applied to automate warehouse operations and improve inventory accuracy.

Code Blame 27 lines (22 loc) · 928 Bytes

```python
from ultralytics import YOLO
import os

# Get absolute path to current script (detect_yolo.py)
base_dir = os.path.dirname(os.path.dirname(__file__))  # Go up one level to project root
model_path = os.path.join(base_dir, "yolo_module", "best.pt")

# Check and load
if not os.path.exists(model_path):
    raise FileNotFoundError(f"'best.pt' not found at: {model_path}")

model = YOLO(model_path)  # ✅ Proper model object

def detect_boxes(frame):
    """
    Accepts a BGR image frame.
    Returns a list of bounding boxes in (x, y, w, h, class_id, confidence) format.
    """
    results = model.predict(source=frame, conf=0.5, verbose=False)

    boxes = []
    for r in results:
        for box, cls_id, conf in zip(r.boxes.xyxy, r.boxes.cls, r.boxes.conf):
            x1, y1, x2, y2 = map(int, box[:4])
            w, h = x2 - x1, y2 - y1
            boxes.append((x1, y1, w, h, int(cls_id), float(conf)))
    return boxes
```

Figure 3: Code Snippet

5. TESTING / VALIDATION

5.1 Overview

The testing and validation process was conducted to ensure that the system performed accurately, efficiently, and reliably across different operating conditions. Multiple testing strategies, including **unit testing**, **integration testing**, and **performance validation**, were used to verify that the system met all design and functional requirements.

5.2 Unit Testing

Unit testing was performed on individual components such as the image capture module, detection logic, and graphical interface. The image capture module was tested for live webcam connectivity and frame retrieval, the YOLOv8 detection module was validated for correct bounding box predictions and box count accuracy, and the Tkinter GUI was tested for button functionality, error handling, and responsiveness. These tests ensured that each module worked correctly before integration.

5.3 Integration Testing

Once all modules were verified individually, integration testing was carried out to ensure smooth communication between components. The complete pipeline—from video capture to detection and display—was tested with live camera input and static images. Various warehouse scenarios were simulated, including different lighting conditions and overlapping boxes. Initial tests showed reduced accuracy in low-light conditions, which was resolved by applying **image preprocessing techniques** like brightness and contrast adjustment using OpenCV, improving detection stability and consistency.

5.4 Performance Validation

The trained YOLOv8 model was validated using standard object detection metrics such as
Precision, **Recall**, and **mAP (Mean Average Precision)** at different IoU thresholds (0.5 and 0.5:0.95). The model achieved high mAP scores, confirming strong generalization across varied environments. The real-time performance was also evaluated using live video feed, where the system maintained consistent frame rates and minimal latency, verifying its suitability for real-world deployment.

5.5 User Testing

To assess usability, the Tkinter GUI was tested by team members to ensure intuitive user interaction. Testers used all available functions such as uploading images, starting live detection, and saving outputs. Feedback helped refine the interface layout, improve image display quality, and optimize system responsiveness. The interface was made more userfriendly through layout adjustments and improved image rendering.

5.6 Issue Resolution

During testing, minor issues such as inaccurate detection in dim lighting, occasional frame lag, and slower inference on CPU were identified. These issues were resolved through techniques like image enhancement, adjusting YOLOv8 confidence thresholds, and reducing frame dimensions to optimize processing speed. Post-resolution, the system operated smoothly with consistent detection performance.

5.7 Summary

Through systematic testing and validation, the system was verified to meet all technical and functional requirements. The model demonstrated accurate box detection, stable real-time performance, and a responsive user interface. By conducting multiple levels of testing and addressing identified issues, the final system achieved reliable and precise results, confirming its effectiveness for warehouse automation and real-world application.

6. RESULTS AND ANALYSIS

The internship project focused on developing an automated system to detect and count boxes in warehouse environments using computer vision and deep learning. The final implementation integrated the object detection model with a simple graphical interface, allowing users to upload images or use live camera input for real-time box counting.

6.1 Model Training and Evaluation

During the training phase, multiple object detection models were tested, including **YOLOv8** and **YOLOv11**, to determine the most accurate and efficient approach for the given dataset. After extensive testing, **YOLOv11** was finalized based on its balanced performance in accuracy and inference speed.

The evaluation metrics used were **Precision**, **Recall**, and **mAP (Mean Average Precision)** — which measure the model's ability to correctly identify, detect, and generalize box objects in various conditions.

6.2 Performance Comparison

Model	**Precision (%)**	**Recall (%)**	**mAP@0.5 (%)**	**mAP@0.5:0.95 (%)**
YOLOv8	87.00	81.00	89.00	—
YOLOv11	84.78	81.43	87.82	68.08

From the above results, it can be observed that:

- **YOLOv8** performed slightly better in **precision** and **mAP@0.5**, indicating high detection accuracy for correctly identified boxes.

- **YOLOv11** achieved more consistent results across stricter thresholds (**mAP@0.5:0.95 = 68.08%**) and offered improved generalization under different lighting and stacking conditions.Both models showed strong recall values, confirming that most actual boxes present in the frame were detected successfully.

6.3 Output Visualization

The final system displayed the number of boxes detected on the image frame, with bounding boxes drawn around each identified box.

The interface allowed both **live camera feed** and **image upload** options through the **Tkinterbased GUI**, enabling users to perform detection seamlessly.

FIGURE 6.1: Output with Bounding Boxes and Count

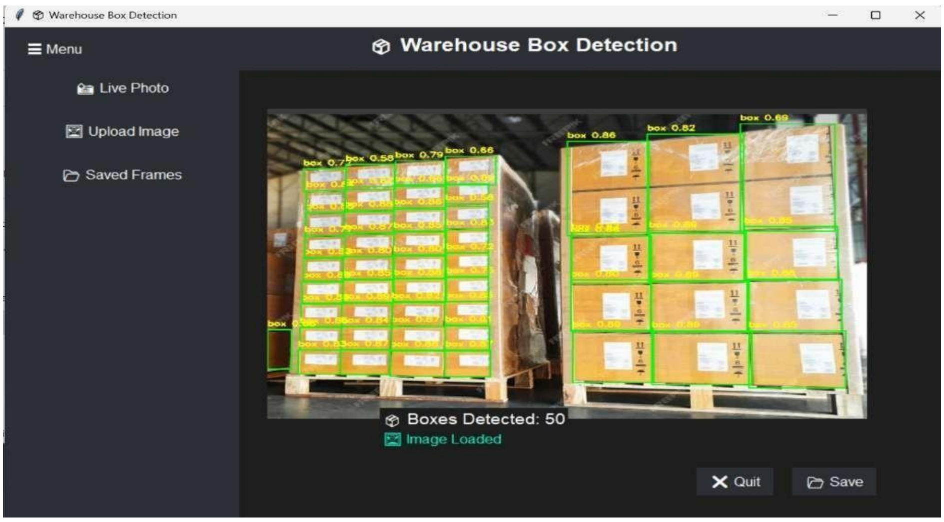

FIGURE 6.2: GUI Window Showing Real-Time Detection

6.4 Result Interpretation

The project successfully demonstrated that computer vision can automate manual counting processes in warehouses with high reliability.

The developed system showed:

- Accurate box detection in diverse environments
- Stable performance in real-time operation
- Easy-to-use interface for non-technical users

The obtained performance metrics validate that the trained YOLOv11 model can be effectively applied for **warehouse inventory automation** and can be further enhanced for industrial-scale deployment.

7. CHALLENGES AND SOLUTIONS

During the development of the project **“Count the Number of Boxes in a Warehouse”**, several technical, operational, and integration-related challenges were encountered. Each challenge presented valuable learning opportunities and helped refine both the technical workflow and understanding of practical AI deployment in industrial environments. The following table elaborates on the major challenges faced and the corresponding solutions implemented:

Challenge	Descrption	Solution/Resolution
1. Dataset Quality and Annotation Issues	Initial dataset images had inconsistent lighting, varying angles, and overlapping boxes, leading to misdetections during training.	The dataset was cleaned and re-annotated using **LabelImg**, ensuring consistent bounding box labeling. Additional samples were collected to improve diversity and robustness.
2. Model Overfitting During Training	The model showed high accuracy on training data but poor performance on validation images.	Techniques like **data augmentation**, **early stopping**, and **dropout regularization** were applied. Learning rate and batch size were tuned for balanced convergence.
3. Real-Time Processing Lag	Real-time detection using webcam input initially caused noticeable lag due to heavy model computation.	Implemented **frame resizing** and used **GPU acceleration** in PyTorch to improve inference speed without sacrificing accuracy.
4. GUI Integration Challenges	Integrating the YOLOv8 detection pipeline with Tkinter GUI caused compatibility issues, especially with OpenCV image formats.	Used **Pillow (PIL)** to convert OpenCV frames into Tkinter-compatible formats and optimized the main loop for smooth frame updates.
5. Model Performance in Cluttered Environments	In warehouse scenes with stacked boxes or partial occlusions, the model occasionally failed to differentiate boundaries.	Trained with more **complex background images** and adjusted **non-max suppression (NMS)** thresholds to better separate overlapping detections.

6. Limited Hardware Resources	Model training on CPU-only systems was slow and memory-intensive.	Shifted training to **Google Colab GPU runtime**, which significantly reduced training time and enabled faster experimentation.
7. Documentation and Reporting	Compiling technical findings, metrics, and workflow diagrams in a professional report required extensive detailing.	Compiling technical findings, metrics, and workflow diagrams in a professional report required extensive detailing.

7.2Learning Outcome:

These challenges strengthened the understanding of real-world computer vision applications. I learned how to handle dataset imbalance, model optimization, integration of AI models with GUI systems, and real-time system deployment. Moreover, it helped in developing a systematic approach toward research, experimentation, and solution refinement.

8. CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

The internship project titled **“Count the Number of Boxes in a Warehouse”** successfully achieved its main objective of designing and implementing an intelligent computer vision system capable of detecting and counting boxes automatically from images and live video feeds. By utilizing the **YOLOv8 and YOLOv11** object detection models, the system was able to provide accurate and efficient box detection results suitable for warehouse automation.

The system integrated multiple technologies — **Python, OpenCV, PyTorch, and Tkinter** — to create a complete and user-friendly desktop application. Through this project, manual labor associated with warehouse inventory counting can be significantly reduced, and the overall efficiency and accuracy of stock management can be improved.

The major accomplishments of this project include:

- Successful detection and counting of boxes in real-time using deep learning.
- Integration of AI-based detection models with an interactive desktop GUI.
- Improved detection performance through optimized training and validation.
- Demonstration of how AI can support automation in industrial and logistics operations.

This project experience provided deep insights into **AI model training, dataset preparation, performance tuning, and interface development**, enhancing both technical and analytical skills. The hands-on exposure to YOLO models and GUI integration bridged theoretical knowledge with practical implementation, which is vital for real-world AI system development.

8.2 Future Scope

While the system performs efficiently under controlled conditions, there is significant scope for improvement and expansion. The following future enhancements are proposed:

1. **Cloud Integration and Centralized Inventory Access:**
   Deploy the application on a **cloud platform** to allow warehouse managers to access and monitor box counts remotely in real time.

2. **Integration with IoT and Warehouse Management Systems (WMS):**

Combine computer vision results with IoT-based inventory sensors or ERP systems for seamless synchronization of warehouse data.

3. **Edge Deployment on Embedded Devices:**
   Optimize the model for deployment on **low-power devices** such as **Raspberry Pi** or **NVIDIA Jetson**, enabling real-time on-site monitoring without dependence on highend systems.

4. **3D Box Counting and Depth Estimation:**
   Integrate **stereo vision** or **depth-sensing cameras** to accurately estimate the number of stacked or partially hidden boxes.

5. **Mobile Application Development:**
   Extend the desktop GUI into a **mobile app** using frameworks like Flutter or React Native, enabling handheld real-time box detection using smartphone cameras.

6. **Multi-Class Object Detection:**
   Enhance the system to detect and classify multiple types of warehouse items such as crates, packages, pallets, and containers.

7. **Continuous Model Improvement via Active Learning:**
   Implement a self-improving mechanism that retrains the model periodically based on newly collected or misclassified images.

The developed project lays a strong foundation for industrial automation using AI and can serve as a prototype for future **smart warehouse management systems** under the Industry 4.0 paradigm.

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